

CHAPTER 2

MASS, WEIGHT AND DENSITY

1.3 Mass and weight

- 1 State that mass is a measure of the quantity of matter in an object at rest relative to the observer
- 2 State that the mass of an object resists change from its state of rest or motion (inertia)
- 3 Know that weights, and therefore masses, may be compared using a beam balance or equal-arm balance
- 4 Describe how to determine mass using an electronic balance
- 5 Describe how to measure weight using a force meter
- 6 Define gravitational field strength as force per unit mass; recall and use the equation

$$\text{gravitational field strength} = \frac{\text{weight}}{\text{mass}}$$

$$g = \frac{W}{m}$$

and know that this is equivalent to the acceleration of free fall

- 7 State that a gravitational field is a region in which a mass experiences a force due to gravitational attraction

1.4 Density

- 1 Define density as mass per unit volume; recall and use the equation

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

$$\rho = \frac{m}{V}$$

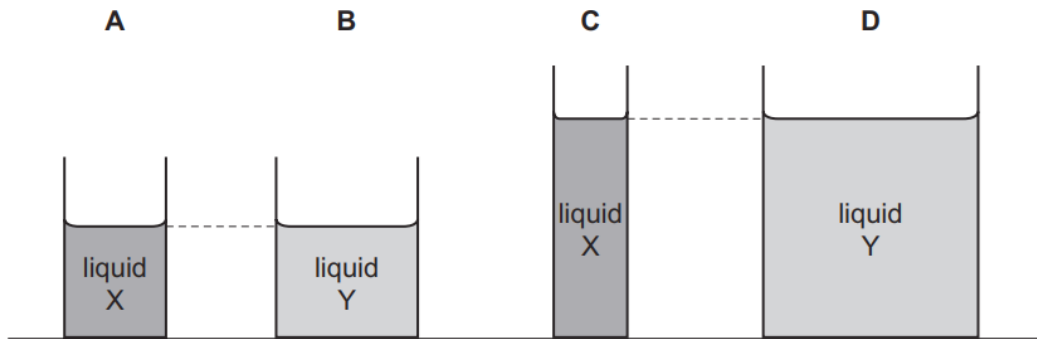
- 2 Describe how to determine the density of a liquid, of a regularly shaped solid and of an irregularly shaped solid which sinks in a liquid (volume by displacement), including appropriate calculations

1. O/N/23/11/Q12

Two liquids, X and Y, are poured into beakers of different sizes.

Liquid X has a density of 1000 kg/m^3 and liquid Y has a density of 900 kg/m^3 .

On the base of which beaker is the pressure greatest?



2. O/N/23/12/Q4

A student finds that the mass of a rock is 65 g.

She pours 30 cm^3 of water into a measuring cylinder.

She places the rock in the water so that the water covers the rock completely. The water level rises to 55 cm^3 .

What is the density of the rock?

- A** 0.38 g/cm^3 **B** 1.2 g/cm^3 **C** 2.2 g/cm^3 **D** 2.6 g/cm^3

3. M/J/23/11/Q5

The gravitational field strength on Earth is 9.8 N/kg .

The gravitational field strength on Mars is 3.7 N/kg .

The difference between the weight of an object on Earth and the weight of the same object on Mars is 25 N.

What is the mass of the object?

- A** 1.9 kg **B** 2.6 kg **C** 4.1 kg **D** 6.8 kg

4. M/J/23/11/Q6

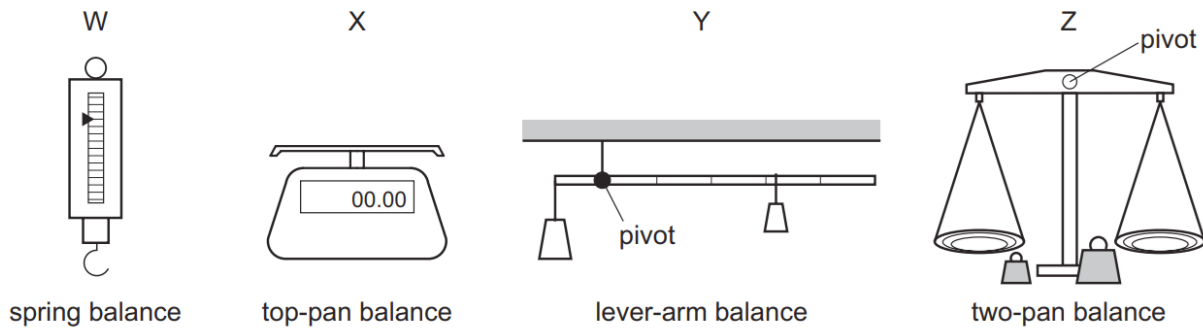
A student determines the density of a solid metal cube.

Which pieces of equipment does the student use?

- A** a balance and a force meter
B a balance and a ruler
C a micrometer and a measuring cylinder
D a ruler and a measuring cylinder

5. M/J/23/12/Q5

5 The diagrams show four balances, W, X, Y and Z.



The scales of all the balances are calibrated on the Earth to measure mass.

Which balances also measure mass correctly when used on the Moon?

- A** W and X **B** W and Z **C** X and Y **D** Y and Z

6. M/J/23/12/Q6

A spacecraft carries a probe to Mars where the gravitational field strength is 3.7 N/kg . The weight of the probe on Mars is 370 N .

The gravitational field strength on Earth is 9.8 N/kg .

What are the mass and the weight of the probe on Earth?

	mass / kg	weight / N
A	100	370
B	100	980
C	370	98
D	370	3700

7. M/J/23/12/Q7

A student determines the density of a liquid.

Which equipment does the student use?

- A** a balance and a ruler
B a balance and a measuring cylinder
C a beaker and a measuring cylinder
D a ruler and a beaker

8. O/N/22/11/Q9

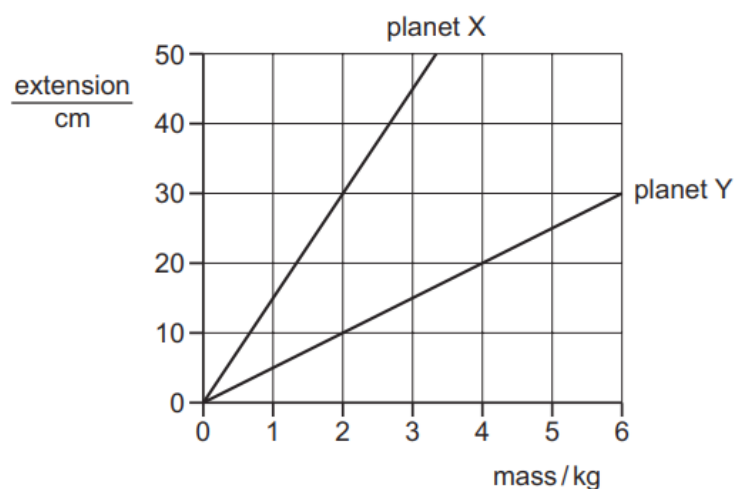
X is the result of Y divided by Z.

What are X, Y and Z?

	X	Y	Z
A	mass	weight	gravitational field strength
B	weight	mass	gravitational field strength
C	mass	gravitational field strength	weight
D	weight	gravitational field strength	mass

9. O/N/22/11/Q10

The graph shows how the extension of a spring changes with the masses suspended from it when the spring is on planet X and when the spring is on planet Y.



Which conclusion can be drawn from these graphs?

- A** It is not possible to compare the gravitational field strengths on planets X and Y.
- B** The gravitational field strength on planet X is equal to the gravitational field strength on planet Y.
- C** The gravitational field strength on planet X is one third of the gravitational field strength on planet Y.
- D** The gravitational field strength on planet X is three times the gravitational field strength on planet Y.

10. O/N/22/12/Q9

A student writes two statements about mass and weight.

- 1 Mass is the property of a body which resists change in its state of rest or motion.
- 2 Weight is the amount of substance in a body.

Which statements are correct?

- A** statement 1 only
B statement 2 only
C statement 1 and statement 2
D neither statement 1 nor statement 2

11. O/N/22/12/Q11

A student finds the density of an irregularly shaped object. He chooses his equipment from this list.

- 1 stop-watch
- 2 measuring cylinder
- 3 balance

Which equipment does the student need to use?

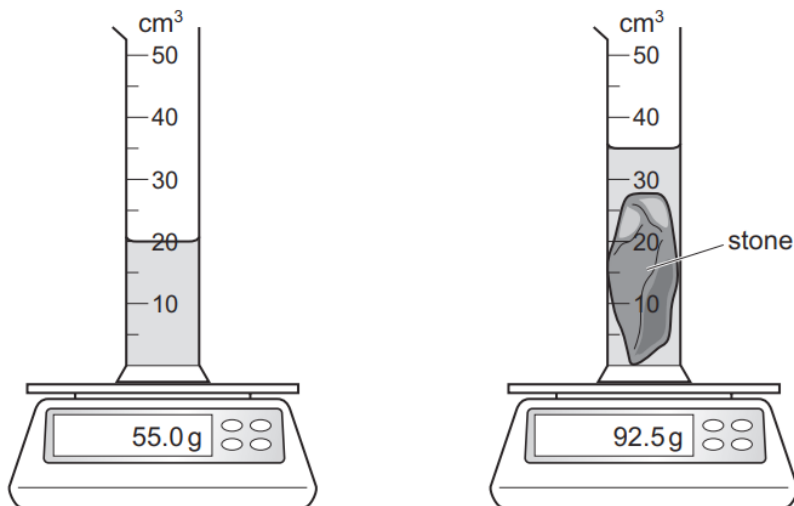
- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

12. M/J/22/12/Q6

A measuring cylinder containing water is placed on a balance.

A stone is placed into the water.

The diagram shows the readings on the balance and on the measuring cylinder.

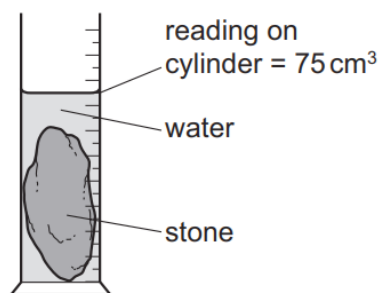


What is the density of the stone?

- A** 1.1 g/cm^3 **B** 1.5 g/cm^3 **C** 2.5 g/cm^3 **D** 2.6 g/cm^3

13. O/N/21/11/Q6

A stone is placed in a measuring cylinder of water, as shown.



The volume of water in the cylinder before adding the stone is 15 cm^3 and the stone has a mass of 90 g.

What is the density of the stone?

- A** 1.0 g/cm^3 **B** 1.2 g/cm^3 **C** 1.5 g/cm^3 **D** 6.0 g/cm^3

14. O/N/21/12/Q6

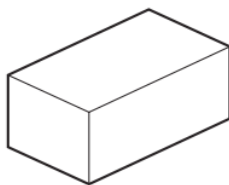
An object in a laboratory is used to determine the gravitational field strength g .

Which quantity is equal to g ?

- A** the weight of the object
B the mass of the object
C the $\frac{\text{weight}}{\text{mass}}$ of the object
D the $\frac{\text{mass}}{\text{weight}}$ of the object

15. O/N/21/12/Q7

A student takes measurements to find the density of a regularly shaped block.



Which formula is used to calculate the density of the block?

- A** $\text{density} = \frac{\text{mass}}{\text{volume}}$
B $\text{density} = \frac{\text{volume}}{\text{mass}}$
C $\text{density} = \frac{\text{weight}}{\text{area}}$
D $\text{density} = \frac{\text{weight}}{\text{volume}}$

16. M/J/21/11/Q6

Which row shows the mass and the weight of an object on the Earth's surface?

[gravitational field strength $g = 10 \text{ N/kg}$]

	mass/kg	weight/N
A	2	0.20
B	2	10
C	5	5.0
D	5	50

17. M/J/21/11/Q7

Water is added to a measuring cylinder containing 100 cm^3 of liquid paraffin.

(The density of paraffin is 0.80 g/cm^3 and that of the water is 1.0 g/cm^3 .)

As the water is added, the level of the paraffin rises to 150 cm^3 . The paraffin and water do not mix.

What finally is the total mass of liquid in the measuring cylinder?

- A** 130 g **B** 140 g **C** 167 g **D** 175 g

18. M/J/21/12/Q6

The gravitational field strength in space is smaller than on the Earth's surface.

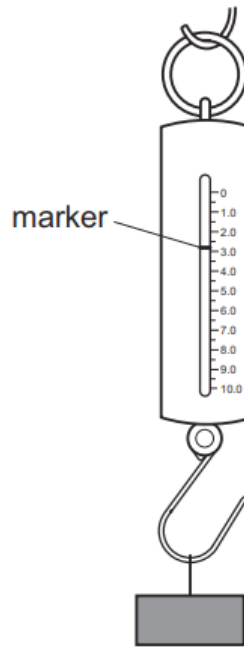
A rocket is used to launch a satellite from the Earth's surface into space.

How are the mass and the weight of the satellite affected as the satellite moves away from the surface of the Earth and into space?

- A** Both the mass and the weight are unaffected.
B The mass decreases and the weight decreases.
C The mass increases and the weight is unaffected.
D The mass is unaffected and the weight decreases.

19. O/N/20/11/Q6

A block is hung on a spring balance. The marker inside the balance is pulled down by the block.



What can the position of the marker be used to determine?

- A** the mass of the block in kg
- B** the mass of the block in N
- C** the moment caused by the block in N
- D** the weight of the block in kg

20. O/N/20/11/Q8

Which statement about mass and weight is correct?

- A** Mass is a scientific term that means the same as weight.
- B** The mass of an object on Earth is 10 times its weight.
- C** Weight is a scalar quantity, mass is a vector quantity.
- D** Weight is the force of gravity pulling on a mass.

21. O/N/20/11/Q9

Which expression is used to calculate density?

- A** mass $\times$ volume
- B** $\frac{\text{mass}}{\text{volume}}$
- C** volume + mass
- D** $\frac{\text{volume}}{\text{mass}}$

22. O/N/20/12/Q8

There is an astronaut on a planet which has a gravitational field strength one fifth of the value on Earth.

He hangs samples of rock from a spring balance designed for use on Earth. Each sample has a mass of 0.1 kg.

The spring balance reads 1.0 kg.

How many samples of rock are there?

- A** 2 **B** 10 **C** 25 **D** 50

23. M/J/20/11/Q9

Four objects of different masses are on different planets.

The weight of each object on its planet is determined.

Which object is on the planet with the smallest gravitational field strength?

	mass / kg	weight / N
A	5	125
B	15	150
C	20	220
D	25	225

24. M/J/20/11/Q10

Which physical quantity does **not** change when a piece of copper is heated?

- A** density
- B** mass
- C** temperature
- D** volume

25. M/J/20/11/Q11

A single metal bolt has a mass of 34 g.

Three of the bolts are immersed in a measuring cylinder that contains 30 cm^3 of water.

The reading on the measuring cylinder rises to 42 cm^3 .

What is the density of the metal?

- A** 0.81 g/cm^3 **B** 2.8 g/cm^3 **C** 7.5 g/cm^3 **D** 8.5 g/cm^3

26. O/N/19/11/Q22

A metal rod is heated.

What happens to the rod?

- A** Its density decreases and its mass decreases.
B Its density decreases and its volume increases.
C Its density increases and its mass decreases.
D Its density increases and its volume increases.

27. O/N/19/12/Q21

A metal disc is heated to 600°C and then lowered into a beaker of water.

What happens to the mass of the metal disc and to the mass of the water in the beaker?

	mass of metal disc	mass of water in beaker
A	increases	decreases
B	increases	no change
C	no change	decreases
D	no change	increases

28. M/J/19/11/Q8

Four objects of different masses are situated in places with different gravitational field strengths.

Which object has the greatest weight?

	mass / kg	<u>gravitational field strength</u> N/kg
A	3.0	10.4
B	3.5	9.5
C	4.0	10.2
D	4.5	9.0

29. M/J/19/12/Q8

The mass of object P is greater than the mass of object Q.

The objects contain different amounts of matter and have a different resistance to change of motion.

Which row is correct?

	greater amount of matter	greater resistance to change of motion
A	P	P
B	P	Q
C	Q	P
D	Q	Q

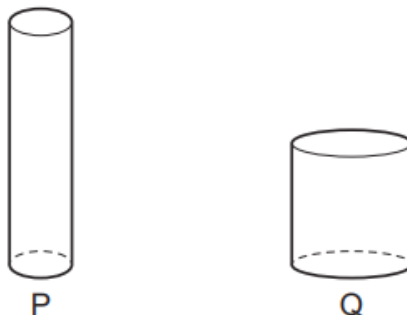
30. O/N/18/12/Q5

Which piece of apparatus may be used to compare the masses of two objects?

- A** balance
- B** manometer
- C** measuring cylinder
- D** micrometer

31. O/N/18/12/Q7

Two cylinders P and Q are made of copper.



The height of P is twice the height of Q. The diameter of P is half the diameter of Q.

Which statement is correct?

- A** The density of cylinder P is four times that of cylinder Q.
- B** The density of cylinder P is twice that of cylinder Q.
- C** The density of cylinder P is equal to that of cylinder Q.
- D** The density of cylinder P is half that of cylinder Q.

32. O/N/18/11/Q6

What happens to an object when it is moved to a location where the gravitational field strength is slightly greater?

- A** Its density decreases.
- B** Its mass decreases.
- C** Its weight increases.
- D** Its volume increases.

33. M/J/18/12/Q9

A body of mass 10 kg falling freely in the gravitational field close to the Moon's surface has an acceleration of 1.6 m/s^2 .

What is the gravitational field strength on the Moon?

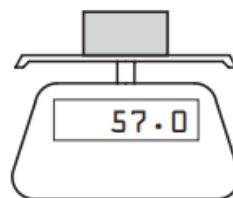
- A** 0 N/kg **B** 1.6 N/kg **C** 10 N/kg **D** 16 N/kg

34. M/J/18/12/Q8

A block of metal is placed on an electronic balance to record its mass.

What is the unit of the reading on the electronic balance and what is the unit of weight?

	unit on electronic balance	unit of weight
A	g	g
B	g	N
C	N	g
D	N	N



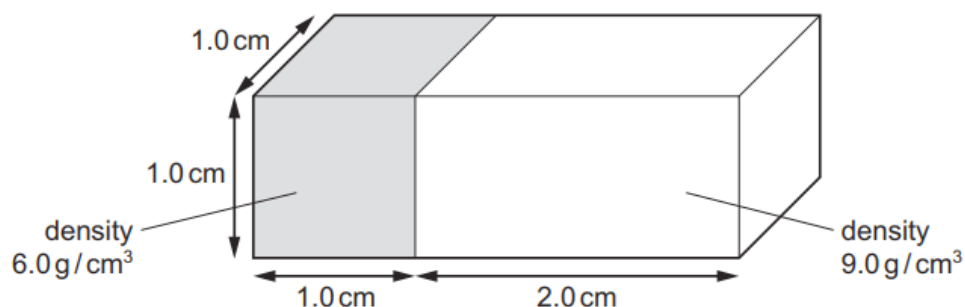
35. M/J/18/11/Q6

Which piece of equipment is used to measure mass?

- A** balance **B** manometer **C** measuring cylinder **D** newton meter

36. O/N/17/12/Q6

Two blocks are joined together.



One block has a density of 6.0 g/cm^3 and the other has a density of 9.0 g/cm^3 .

What is the overall density of the two blocks joined together?

- A** 7.0 g/cm^3 **B** 7.5 g/cm^3 **C** 8.0 g/cm^3 **D** 15 g/cm^3

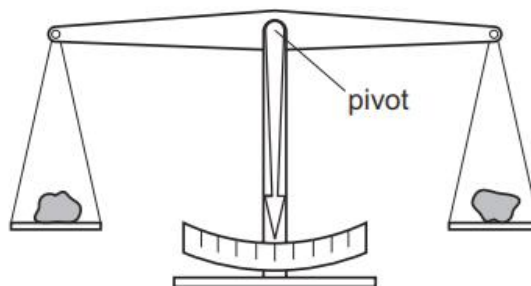
37. O/N/17/12/Q5

The diagram shows two objects on a beam balance.

The beam balance is in equilibrium.

Which quantities may be different?

- A** the masses of the two objects
B the moments about the pivot of the two objects
C the volumes of the two objects
D the weights of the two objects



38. M/J/16/11/Q8

A force acts on a body.

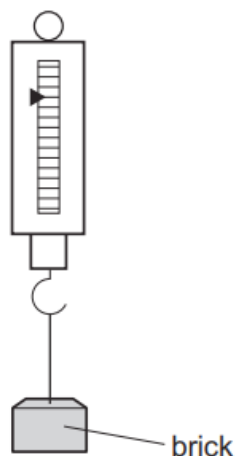
Which list contains only quantities that can be changed by the force?

- A** mass, shape, velocity **C** mass, velocity, volume
B mass, shape, volume **D** shape, velocity, volume

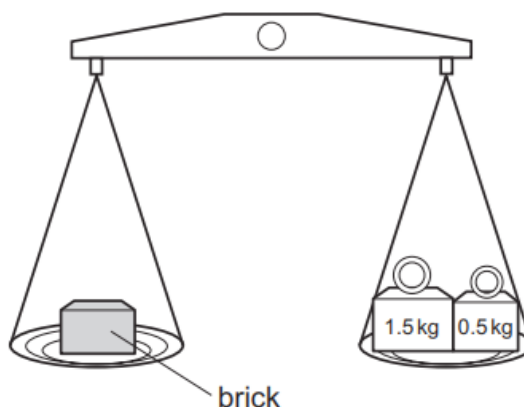
39. M/J/16/12/Q5

A brick is placed on a newton meter and then on a beam balance.

newton meter



beam balance



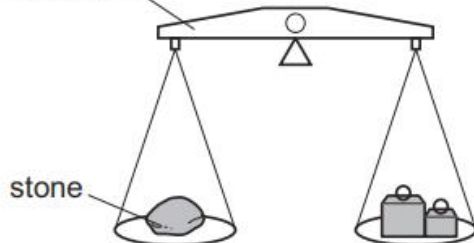
What is measured by each instrument?

	newton meter	beam balance
A	mass	mass
B	mass	weight
C	weight	mass
D	weight	weight

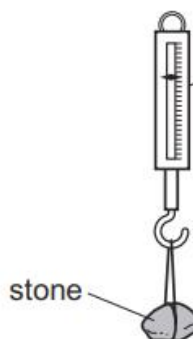
40. M/J/15/12/Q8

The mass of a stone is found on Earth using a pan balance. The weight of the stone is found using a newton meter.

pan balance



newton meter



Are the readings the same or different on the Moon?

	reading on pan balance	reading on newton meter
A	different	different
B	same	different
C	different	same
D	same	same

41. M/J/14/12/Q8

A stone has a mass of 390 g and a density of 2.7 g/cm^3 .

Cooking oil has a density of 0.90 g/cm^3 .

Which mass of oil has the same volume as the stone?

- A** 130 g **B** 160 g **C** 900 g **D** 1200 g

42. M/J/14/11/Q8

Four objects are situated in places with different gravitational field strengths.

Which object has the greatest weight?

	<u>mass</u> kg	<u>gravitational field strength</u> N/kg
A	3.0	10.4
B	3.5	9.5
C	4.0	10.2
D	4.5	9.0

43. M/J/13/11/Q6

Paper is sold in packets labelled 80 g/m^2 . This means that a sheet of paper of area 10000 cm^2 has a mass of 80 g.

The thickness of each sheet is 0.11 mm.

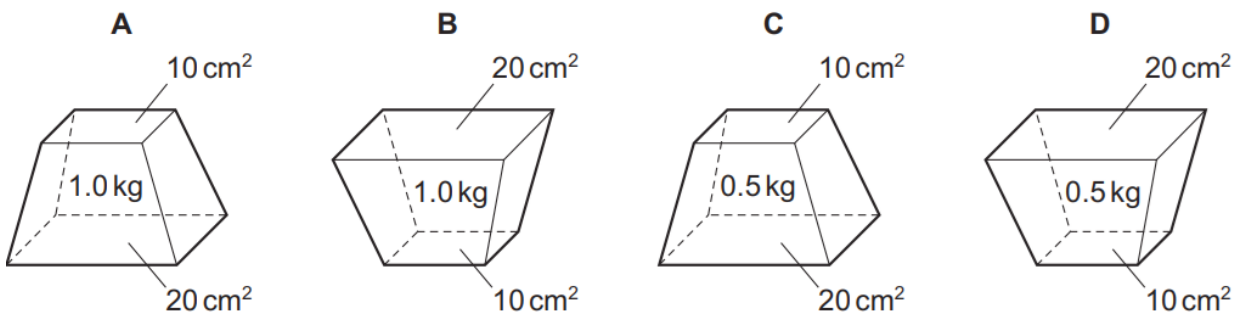
What is the density of the paper?

- A** 0.073 g/cm^3 **B** 0.088 g/cm^3 **C** 0.73 g/cm^3 **D** 0.88 g/cm^3

44. O/N/12/12/Q11

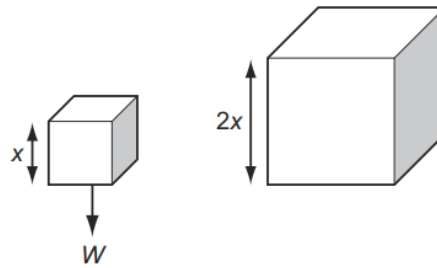
Four solid blocks with flat surfaces are placed on some soft plasticine, in the positions shown.

Which block sinks the deepest into the plasticine?



45. O/N/12/11/Q6

The diagram shows two cubes made from the same material. One cube has sides that are twice as long as the sides of the other cube.



The weight of the small cube is W .
What is the weight of the larger cube?

- A** $2W$ **B** $4W$ **C** $8W$ **D** $16W$

46. O/N/12/12/Q6

The mass and the volume of a bar made from metal X are measured.
The masses and volumes of four other bars are measured.
Which bar is made from a metal with a density that is double that of X?

	mass compared with X	volume compared with X
A	double	half
B	half	same
C	same	double
D	same	half

47. M/J/12/11/Q10

A block of metal is taken from the Earth to the Moon. Which property of the block changes?

- A** density **C** volume
B mass **D** weight

48. M/J/11/12/Q15

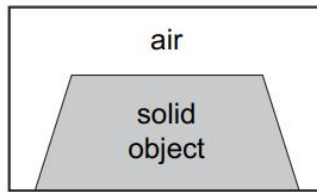
A balloon filled with air is gently heated.
What happens to the mass and the density of the air inside the balloon?

	mass	density
A	decreases	decreases
B	decreases	stays the same
C	stays the same	decreases
D	stays the same	stays the same

49. O/N/10/12/Q11

A box has an internal volume of 1000 cm^3 . When a solid object is placed in the closed box, the volume of air in the box is 520 cm^3 .

The density of the object is 8.00 g/cm^3 .



What is the mass of the object?

- A** 60.0g **B** 3840g **C** 4160g **D** 8000g

50. O/N/10/12/Q9

A person of weight 600 N at the bottom of a mountain climbs to the top. The gravitational field strength changes from 10.00 N/kg at the bottom to 9.97 N/kg at the top. His mass is unchanged as he climbs.

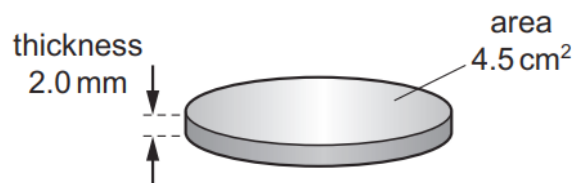
What are his mass and his weight at the top of the mountain?

	mass at top of mountain / kg	weight at top of mountain / N
A	60.0	598
B	60.0	600
C	60.1	598
D	60.1	600

51. O/N/16/12/Q8

The density of gold is 19 g/cm^3 .

A gold coin is 2.0 mm thick. Each face of the coin has an area of 4.5 cm^2 .



What is the mass of the coin?

- A** 2.1g **B** 17g **C** 21g **D** 170g

52. M/J/10/12/Q6

A passenger is sitting in an aeroplane, which takes off and climbs to 10 000 m.

During this time, what happens to the mass and to the weight of the passenger?

	mass	weight
A	decreases	decreases
B	increases	increases
C	unchanged	decreases
D	unchanged	increases

53. O/N/09/Q7

A room measures $4.0 \text{ m} \times 3.0 \text{ m} \times 2.0 \text{ m}$. The density of the air in the room is 1.3 kg/m^3 .

What is the mass of air in the room?

- A** 0.054 kg **B** 18 kg **C** 24 kg **D** 31 kg

54. M/J/09/Q7

A measuring cylinder contains 118 cm^3 of water. When a small object is fully immersed in the water, the reading goes up to 132 cm^3 . The object has a mass of 42 g.

What is the density of the object?

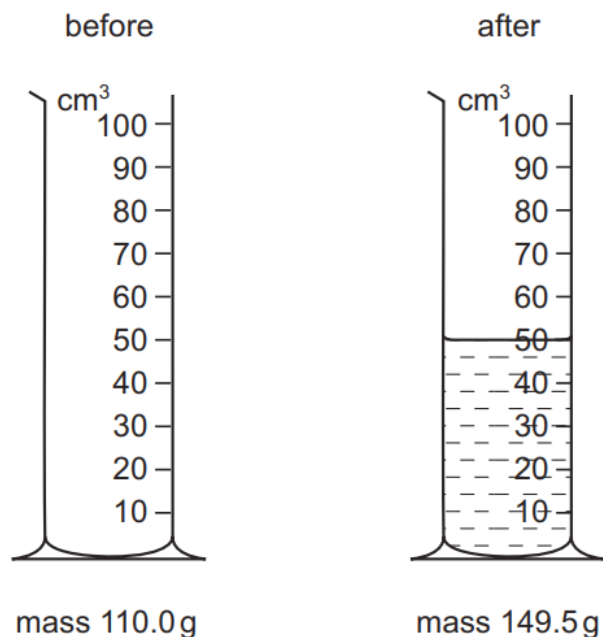
- A** $\frac{14}{42} \text{ g/cm}^3$ **B** $\frac{42}{14} \text{ g/cm}^3$ **C** $\frac{42}{118} \text{ g/cm}^3$ **D** $\frac{132}{42} \text{ g/cm}^3$

55. O/N/08/Q6

The mass of a measuring cylinder is measured before and after pouring a liquid into it.

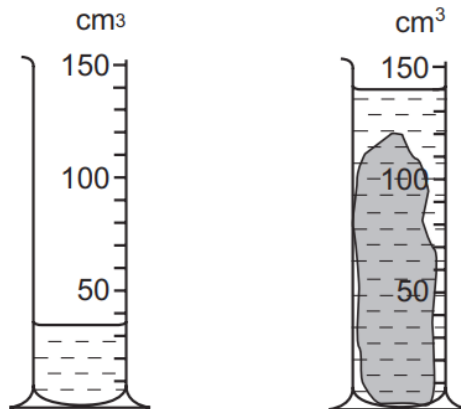
What is the density of the liquid?

- A** 0.79 g/cm^3
B 1.3 g/cm^3
C 1.4 g/cm^3
D 2.2 g/cm^3



56. M/J/08/Q6

A lump of metal has a mass of 210 g. It is lowered into a measuring cylinder containing water. The level of the water rises from 35 cm^3 to 140 cm^3 .

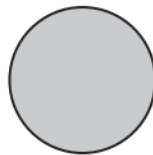


What is the density of the metal?

- A** 0.67 g/cm^3 **B** 1.5 g/cm^3 **C** 2.0 g/cm^3 **D** 6.0 g/cm^3

57. O/N/06/Q6

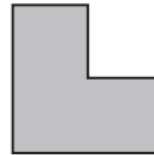
Three objects are cut from the same sheet of steel. They are different shapes but they all have the same mass.



disc



square



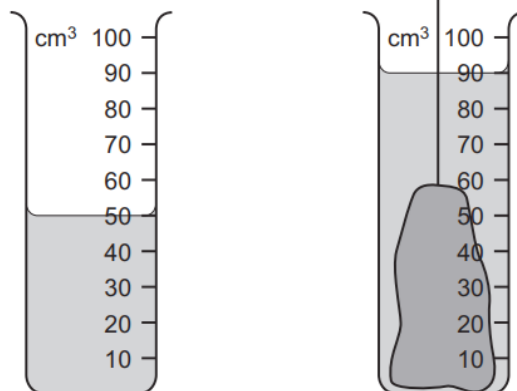
L-shape

Which object has the greatest density?

- A** the disc **B** the L-shape **C** the square **D** they all have the same density

58. M/J/06/Q7

An object of mass 100 g is immersed in water as shown in the diagram.

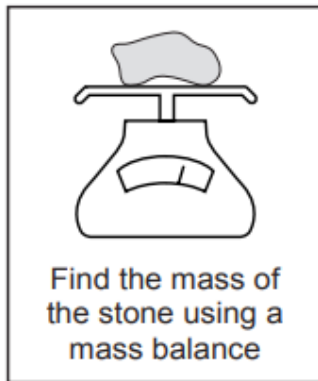


What is the density of the material from which the object is made?

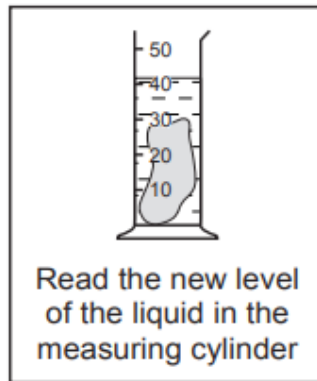
- A** 0.4 g/cm^3 **B** 0.9 g/cm^3 **C** 1.1 g/cm^3 **D** 2.5 g/cm^3

59. O/N/05/Q7

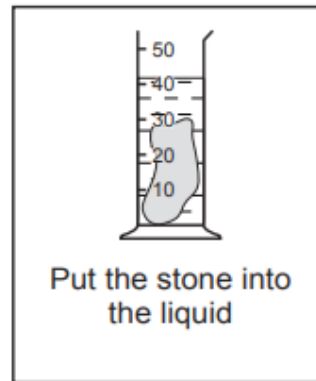
A student is trying to find the density of a stone, but he has mixed up the instruction cards.



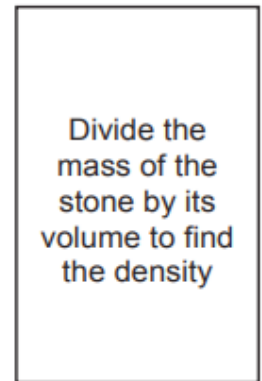
Card 1



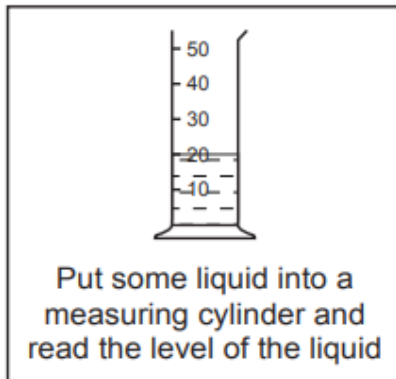
Card 2



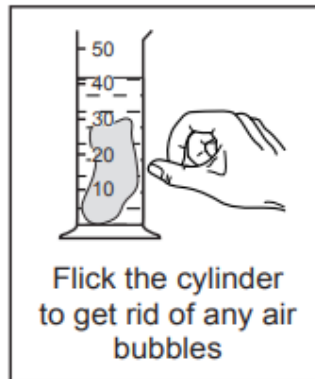
Card 3



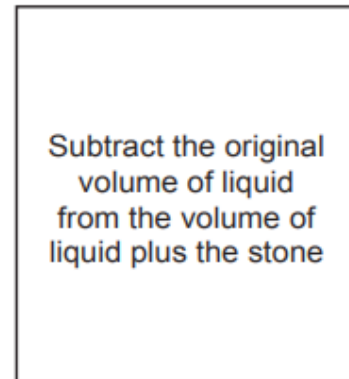
Card 4



Card 5



Card 6



Card 7

What order should the cards be in?

A 5 → 3 → 6 → 2 → 1 → 4 → 7

C 5 → 6 → 3 → 2 → 1 → 7 → 4

B 1 → 5 → 3 → 6 → 2 → 7 → 4

D 1 → 4 → 5 → 3 → 6 → 2 → 7

60. O/N/05/Q6

An object that has a mass of 15 kg on the Earth is taken to the Moon.

The gravitational field strength on the Earth is 10 N/kg and on the Moon is 1.6 N/kg.

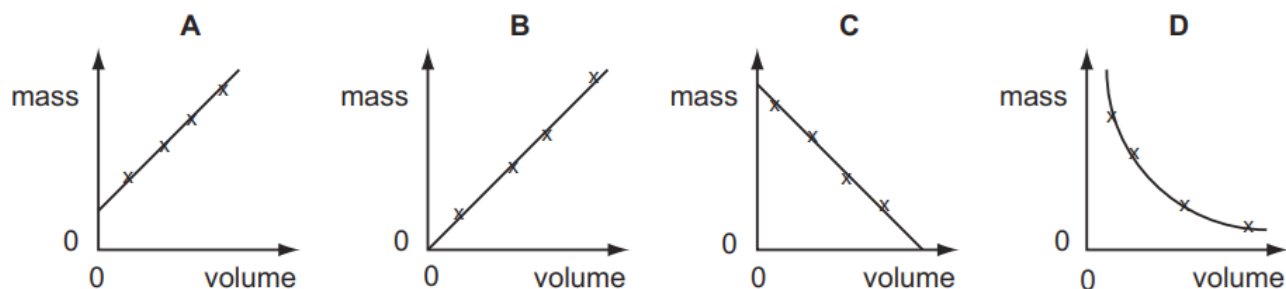
What are the mass and the weight of the object on the Moon?

	mass / kg	weight / N
A	15	24
B	15	150
C	24	15
D	150	24

61. M/J/05/Q5

Some students measure the masses and the volumes of different sized samples of a type of wood.

Which graph shows their results?



62. M/J/05/Q4

At a point on the surface of the Earth, the gravitational field strength is 9.8 N/kg .

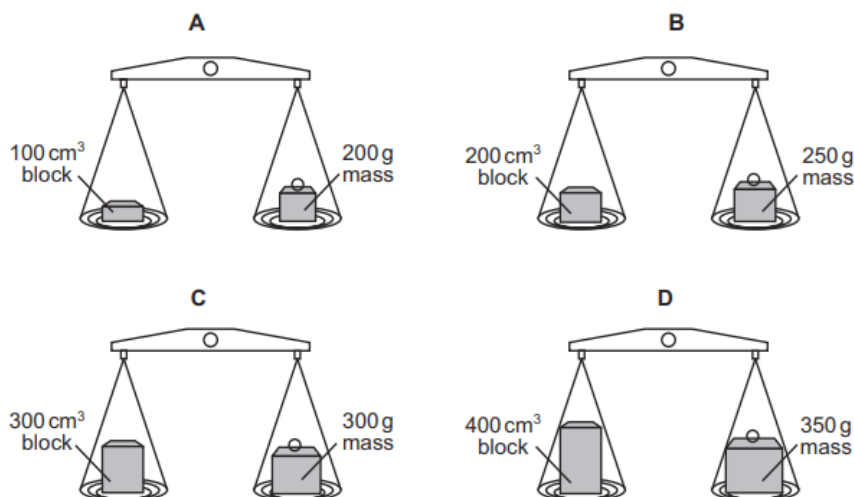
Which pair of values for mass and weight are correct for an object placed at this point?

	mass / kg	weight / N
A	9.8	10
B	10	9.8
C	10	98
D	98	10

63. O/N/04/Q7

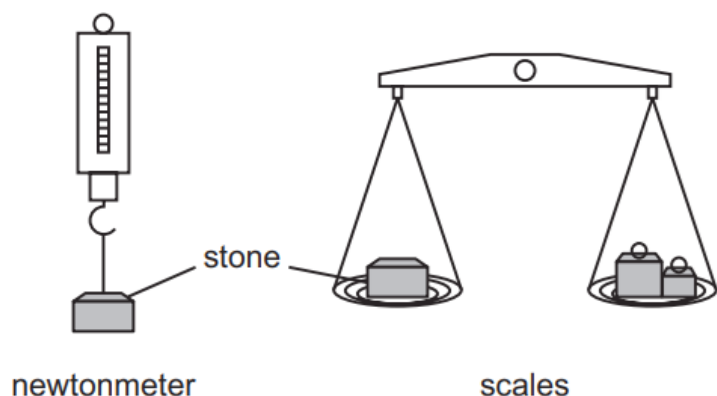
Four blocks, each made from a different material, are placed on scales and balanced as shown in the diagrams below.

In which diagram does the block have the greatest density?



64. O/N/04/Q6

A stone is weighed using a newtonmeter (spring balance) and a pair of scales (pan balance).



This experiment is repeated on the Moon.

Are the readings for each balance the same or different when taken on Earth and on the Moon?

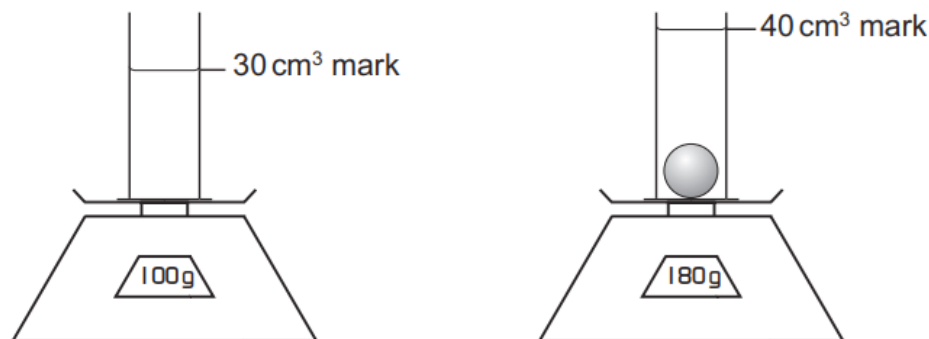
	reading on newtonmeter	reading on scales
A	different	different
B	different	same
C	same	different
D	same	same

65. M/J/04/Q5

A measuring cylinder containing some water stands on a scale pan. A solid ball is lowered into the water.

The water level rises from the 30 cm^3 mark to the 40 cm^3 mark.

The scale reading increases from 100 g to 180 g.



What is the density of the material of the ball?

- A** 2.0 g/cm^3 **B** 4.5 g/cm^3 **C** 8.0 g/cm^3 **D** 18 g/cm^3

66. M/J/04/Q4

Which relationship defines gravitational field strength?

- A** mass x 10 **B** mass x weight **C** mass/weight **D** weight/mass

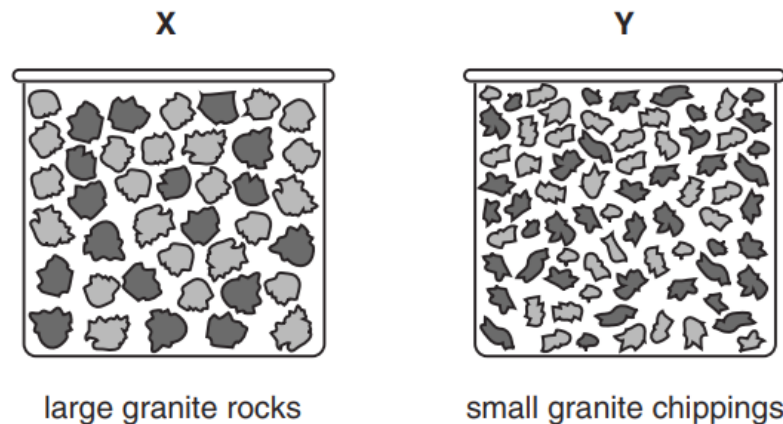
67. O/N/03/Q6

Which property of a body **cannot** be changed if a force is applied to it?

- A** its mass **B** its shape **C** its size **D** its velocity

68. O/N/03/Q5

A box **X** full of large granite rocks is weighed. An identical box **Y** full of small granite chippings is then weighed.



Which box weighs more and why?

	heavier box	reason
A	X	there is more air in box X
B	X	the density of a chipping is less than a rock
C	Y	there is less air in box Y
D	Y	the density of a chipping is greater than a rock

69. M/J/14/12/Q11

A force is applied to a body.

Which property of the body **cannot** be changed by the force?

- A** its mass **B** its shape **C** its size **D** its velocity

70. M/J/14/12/Q10

A man uses clay to make a pot. He wants the pot to be as stable as possible when placed on a flat surface.

Which two features of the pot must the man consider?

- A** the area of the base and the height of the centre of gravity
- B** the density of the clay and the area of the base
- C** the density of the clay and the height of the centre of gravity
- D** the weight and the height of the centre of gravity

71. M/J/03/Q4

Ten identical steel balls, each of mass 27 g, are immersed in a measuring cylinder containing 20 cm^3 of water.

The reading of the water level rises to 50 cm^3 .

What is the density of the steel?

- A** 0.90 g/cm^3 **B** 8.1 g/cm^3 **C** 9.0 g/cm^3 **D** 13.5 g/cm^3

72. M/J/03/Q3

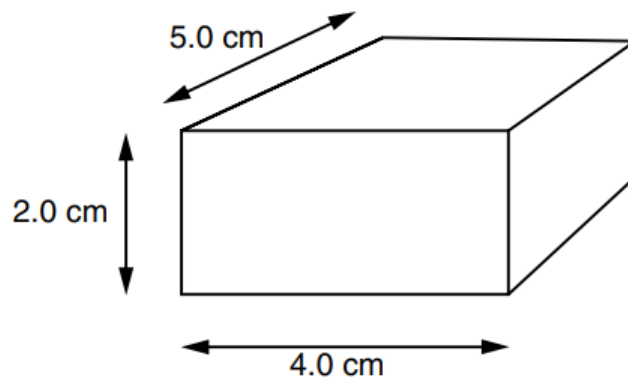
On the Earth, the gravitational field strength is 10 N/kg . On the Moon, the gravitational field strength is 1.6 N/kg .

If an object has a weight of 50 N on Earth, what is its weight on the Moon?

- A** 1.6 N **B** 5.0 N **C** 8.0 N **D** 80 N

73. O/N/02/Q5

The diagram shows a material with dimensions $5 \text{ cm} \times 4 \text{ cm} \times 2 \text{ cm}$. It has a mass of 100 g .



What is the density of the material?

- A** 0.40 g/cm^3 **B** 2.5 g/cm^3 **C** 5.0 g/cm^3 **D** 10 g/cm^3